

9.7 Polynomial Matrices and the Smith Normal Form

The rational canonical form of the previous section is a complete answer to the similarity problem, but the proof did not tell us how to *find* the invariant factors of a given matrix. This section supplies the missing algorithm, and it comes from an unexpected place: instead of working with $\mathbf{A}$, we work with the single polynomial matrix $x\mathbf{I} - \mathbf{A}$, and we reduce it by row and column operations whose scalars are allowed to be polynomials. The reduced form is diagonal, its non-constant diagonal entries are the invariant factors, and two matrices are similar exactly when their reduced forms agree. On the way we finally pay off a promise from Chapter 3: every square matrix is similar to its transpose, over every field.

9.7.1 The space as a module over the polynomials

Everything in this chapter is built by applying **polynomials** in T to vectors: the T -annihilator of a vector, the cyclic subspace $Z(\mathbf{v}; T)$, the kernels $\ker p(T)$. Name that structure once. For V a vector space over F and $T \in \mathcal{L}(V)$, the product

$$F[x] \times V \rightarrow V, \quad p \cdot \mathbf{v} := p(T) \mathbf{v}$$

satisfies, by Theorem 5.7.1, exactly the vector space axioms with F replaced by $F[x]$: it is additive in each factor, $(pq) \cdot \mathbf{v} = p \cdot (q \cdot \mathbf{v})$, and $1 \cdot \mathbf{v} = \mathbf{v}$. A set with an addition and a product by the elements of a commutative ring R obeying these rules is an **R -module**, so the pair (V, T) is the same thing as an $F[x]$ -module that is finite-dimensional over the constants F . We develop no general theory of modules; the point is the dictionary:

$F[x]$ -module language	operator language
submodule of V	T -invariant subspace (Definition 8.1.1)
submodule generated by $\mathbf{v}$, that is $F[x] \cdot \mathbf{v}$	cyclic subspace $Z(\mathbf{v}; T)$ (Definition 9.5.1)
the annihilator of $\mathbf{v}$ in $F[x]$	the T -annihilator of $\mathbf{v}$
V is generated by one element	T has a cyclic vector

A subspace closed under multiplication by x is closed under multiplication by every polynomial, which is why the first two lines match.

Where does $x\mathbf{I} - \mathbf{A}$ come from here? Fix $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ with $\mathbf{A} = [T]_{\mathcal{B}}$. As an $F[x]$ -module, V is generated by $\mathbf{b}_1, \dots, \mathbf{b}_n$, subject to the n relations $x \cdot \mathbf{b}_j = \sum_i a_{ij} \mathbf{b}_i$, whose coefficient vectors are the columns of $x\mathbf{I} - \mathbf{A}$. So $x\mathbf{I} - \mathbf{A}$ is the **matrix of relations**, and changing generators or replacing one relation by a combination of relations should not change V . That is the intuition behind the operations we now allow; the proofs below are matrix proofs and do not rely on it.

9.7.2 Row and column operations with polynomial multipliers

A matrix in $M_n(F[x])$ will be called a **polynomial matrix**. Over a field we reduced a matrix by three kinds of row operation (Definition 2.2.1). Over $F[x]$ we must be careful with one of

them: multiplying a row by x cannot be undone, because x has no inverse in $F[x]$. So the scaling operation keeps its scalar in F , while the operation that adds a multiple of one row to another may use any polynomial multiplier.

Definition 9.7.1 — Elementary Operations over $F[x]$

Let $\mathbf{M} \in M_n(F[x])$. An **elementary row operation** over $F[x]$ on $\mathbf{M}$ is one of:

- (E1) interchanging two rows;
- (E2) multiplying one row by a **non-zero scalar** $c \in F$;
- (E3) adding h times one row to a **different** row, where $h \in F[x]$ is **any** polynomial.

Elementary column operations over $F[x]$ are defined the same way with “column” in place of “row”.

In words: (E1) and (E3) are as over a field, except that (E3) now permits a polynomial multiplier; (E2) is restricted, because only the non-zero constants have inverses in $F[x]$. Each of the three is reversible by an operation of the same kind: swap back, scale by c^{-1} , add $-h$ times the row back.

Two polynomial matrices reachable from one another by such operations should count as the same. As over a field (Definition 3.8.1), the clean way to say this is with invertible matrices on both sides — but “invertible” must now mean invertible *inside* $M_n(F[x])$.

Proposition 9.7.1 — Invertible Polynomial Matrices

Let $\mathbf{U} \in M_n(F[x])$. Then $\mathbf{U}$ has an inverse in $M_n(F[x])$ if and only if $\det \mathbf{U}$ is a **non-zero constant**.

Proof. ($\Rightarrow$) Suppose $\mathbf{U}\mathbf{W} = \mathbf{I}$ with $\mathbf{W} \in M_n(F[x])$. Taking determinants and using multiplicativity over the commutative ring $F[x]$ (Theorem 6.5.4 and the remark on commutative rings after it), $\det \mathbf{U} \cdot \det \mathbf{W} = 1$. By Theorem 0.8.3, $\deg(\det \mathbf{U}) + \deg(\det \mathbf{W}) = \deg 1 = 0$, and degrees of non-zero polynomials are ≥ 0 , so $\deg(\det \mathbf{U}) = 0$; that is, $\det \mathbf{U} \in F \setminus \{0\}$.

($\Leftarrow$) Suppose $\det \mathbf{U} = c \in F \setminus \{0\}$. The adjugate identity

$$\mathbf{U} \operatorname{adj} \mathbf{U} = \operatorname{adj} \mathbf{U} \mathbf{U} = (\det \mathbf{U}) \mathbf{I}$$

holds over any commutative ring (Theorem 6.6.2 and the remark on commutative rings following it), and $\operatorname{adj} \mathbf{U} \in M_n(F[x])$, since its entries are minors of $\mathbf{U}$. So $\mathbf{W} = c^{-1} \operatorname{adj} \mathbf{U} \in M_n(F[x])$ satisfies $\mathbf{U}\mathbf{W} = \mathbf{W}\mathbf{U} = \mathbf{I}$. ■

Definition 9.7.2 — Equivalence over $F[x]$

Let $\mathbf{M}, \mathbf{M}' \in M_n(F[x])$. We say $\mathbf{M}$ and $\mathbf{M}'$ are **equivalent** over $F[x]$, written $\mathbf{M} \approx \mathbf{M}'$,

if there are $\mathbf{U}, \mathbf{V} \in M_n(F[x])$, each invertible in $M_n(F[x])$, with

$$\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}.$$

Proposition 9.7.2 — Operations and Equivalence

Let $\mathbf{M}, \mathbf{M}' \in M_n(F[x])$.

- (a) $\approx$ is an equivalence relation on $M_n(F[x])$.
- (b) If $\mathbf{M}'$ is obtained from $\mathbf{M}$ by a finite sequence of elementary row and column operations over $F[x]$, then $\mathbf{M} \approx \mathbf{M}'$.

Proof.

- (a) Take $\mathbf{U} = \mathbf{V} = \mathbf{I}$ for reflexivity. If $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$, then $\mathbf{M} = \mathbf{U}^{-1}\mathbf{M}'\mathbf{V}^{-1}$, and $\mathbf{U}^{-1}, \mathbf{V}^{-1}$ are invertible in $M_n(F[x])$; this gives symmetry. If $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$ and $\mathbf{M}'' = \mathbf{U}'\mathbf{M}'\mathbf{V}'$, then $\mathbf{M}'' = (\mathbf{U}'\mathbf{U})\mathbf{M}(\mathbf{V}\mathbf{V}')$, and a product of invertible matrices is invertible, which gives transitivity.
- (b) Each elementary row operation over $F[x]$ is left multiplication by the matrix obtained by performing that operation on $\mathbf{I}_n$, exactly as over a field (Theorem 2.4.1, whose proof uses only the ring operations). Those three matrices are $\mathbf{P}_{ij}$, $\mathbf{D}_i(c)$ and $\mathbf{I} + h\mathbf{E}_{ij}$ with $i \neq j$, of determinants -1 , $c \neq 0$ and 1 respectively (Theorem 6.4.6 and Theorem 6.5.1, valid over a commutative ring), hence invertible in $M_n(F[x])$ by Proposition 9.7.1. Column operations are right multiplication by the same matrices. Now use (a).

■

Warning. Multiplying a row by a non-constant polynomial is not an allowed operation. It changes the answer. The matrix $\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ becomes $\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}$ after “multiplying row 1 by x ”, and the two are not equivalent: their determinants are 1 and x , and by Proposition 9.7.1 the first is invertible over $F[x]$ while the second is not. The rule to remember is that only the **units** of $F[x]$ — the non-zero constants — may be used as scaling factors.

9.7.3 The Smith normal form

Every polynomial matrix can be brought to a diagonal form whose diagonal entries divide one another in turn, and that form is unique.

The existence proof is Euclidean division, run on whichever non-zero entry has the smallest degree: if that entry fails to divide one of its neighbors in the same row or column, the remainder gives a non-zero entry of strictly smaller degree, and degrees cannot decrease forever.

Theorem 9.7.3 — Smith Normal Form

Let F be a field, $n \geq 1$ and $\mathbf{M} \in M_n(F[x])$. Then:

- (a) **(Existence)** $\mathbf{M}$ is equivalent over $F[x]$, by a finite sequence of elementary row and column operations, to a diagonal matrix

$$\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$$

in which every d_i is **monic or zero** and $d_i \mid d_{i+1}$ for $i = 1, \dots, n-1$.

- (b) **(Uniqueness)** The list $(d_1, \dots, d_n)$ is determined by $\mathbf{M}$: if $\mathbf{M} \approx \text{diag}(d_1, \dots, d_n) \approx \text{diag}(e_1, \dots, e_n)$ with both lists as in (a), then $d_i = e_i$ for every i .

The matrix $\mathbf{D}$ is called the **Smith normal form** of $\mathbf{M}$, and $d_1, \dots, d_n$ its **invariant divisors**.

Since $0 \mid f$ holds only for $f = 0$, the divisibility chain forces the zero entries, if any, to be the last ones.

Idea. Step roadmap for existence. ① Among all matrices reachable from $\mathbf{M}$ by operations, pick one whose smallest non-zero entry degree is least, and move that entry to the corner. ② That corner entry f must divide every other entry of its row and column: otherwise divide with remainder and produce an entry of smaller degree, contradicting the choice in ①. ③ Clear the first row and column with f ; the same remainder trick, after adding a row to the first one, shows f divides every entry of what is left. ④ Induct on the size. The divisibility $d_1 \mid d_2$ survives the induction because operations replace entries by $F[x]$ -combinations of entries.

For uniqueness, find a quantity that can be read off both $\mathbf{M}$ and $\mathbf{D}$. The greatest common divisor of all $k \times k$ minors is such a quantity: minors of a product are $F[x]$ -combinations of minors of the factors (Cauchy–Binet), so the gcd cannot change under equivalence, and for a diagonal matrix with a divisibility chain it is simply $d_1 d_2 \cdots d_k$.

Proof.

- (a) If $\mathbf{M} = 0$, it is already of the stated form. So assume $\mathbf{M} \neq 0$.
For a non-zero $\mathbf{N} \in M_n(F[x])$ put

$$\delta(\mathbf{N}) := \min\{\deg n_{ij} : n_{ij} \neq 0\} \in \mathbb{N},$$

the least degree of a non-zero entry. Let $\mathcal{S}$ be the set of all non-zero matrices obtainable from $\mathbf{M}$ by finite sequences of elementary row and column operations over $F[x]$; it contains $\mathbf{M}$. The set $\{\delta(\mathbf{N}) : \mathbf{N} \in \mathcal{S}\}$ is a non-empty subset of $\mathbb{N}$, so it has a least element; **fix $\mathbf{N} \in \mathcal{S}$ with $\delta(\mathbf{N})$ least.**

Step 1. Put a least-degree entry in the corner, monic. Choose (i, j) with $n_{ij} \neq 0$ and $\deg n_{ij} = \delta(\mathbf{N})$. Swapping row i with row 1 and column j with column 1 (operations (E1)), and then scaling the new first row by the inverse of the leading coefficient of the corner entry (operation (E2)), we obtain a matrix in $\mathcal{S}$ whose $(1, 1)$ -entry f is **monic** of degree $\delta(\mathbf{N})$. These operations permute the entries and rescale some of them, so the new matrix still has least non-zero entry degree $\delta(\mathbf{N})$. Rename it $\mathbf{N}$.

Step 2. f divides every entry of the first row and the first column. Suppose $f \nmid n_{1j}$ for some $j \geq 2$. By Theorem 5.1.1, $n_{1j} = qf + r$ with $r \neq 0$ and $\deg r < \deg f$. Adding $-q$ times column 1 to column j (operation (E3)) produces a matrix $\mathbf{N}' \in \mathcal{S}$ whose $(1, j)$ -entry is $r \neq 0$ with $\deg r < \deg f = \delta(\mathbf{N})$. Then

$\delta(\mathbf{N}') < \delta(\mathbf{N})$, contradicting the minimality of $\delta(\mathbf{N})$. Hence $f \mid n_{1j}$ for every j , and the same argument with rows in place of columns gives $f \mid n_{i1}$ for every i .

Step 3. Clear the first row and column. For $j \geq 2$ add $-(n_{1j}/f)$ times column 1 to column j ; the multipliers are polynomials by Step 2, and after these operations the first row is $(f, 0, \dots, 0)$ while the first column is unchanged. Then for $i \geq 2$ add $-(n_{i1}/f)$ times row 1 to row i ; since the first row is now $(f, 0, \dots, 0)$, this clears the first column and changes nothing else. The result is

$$\begin{pmatrix} f & 0 \\ 0 & \mathbf{M}' \end{pmatrix} \in \mathcal{S}, \quad \mathbf{M}' \in M_{n-1}(F[x]),$$

and its least non-zero entry degree is still $\delta(\mathbf{N})$: it is $\geq \delta(\mathbf{N})$ by minimality, and f is an entry of degree $\delta(\mathbf{N})$.

Step 4. f divides every entry of $\mathbf{M}'$. Suppose $f \nmid (\mathbf{M}')_{kl}$ for some k, l . Add row $k+1$ to row 1. The new first row is $(f, (\mathbf{M}')_{k1}, \dots, (\mathbf{M}')_{k,n-1})$, so the $(1, l+1)$ -entry of the resulting matrix $\mathbf{N}'' \in \mathcal{S}$ is $(\mathbf{M}')_{kl}$, which is not divisible by f , while the $(1, 1)$ -entry is still f , monic of degree $\delta(\mathbf{N})$. By minimality $\delta(\mathbf{N}'') = \delta(\mathbf{N})$, so Step 2 applies to $\mathbf{N}''$ and says that f divides every entry of the first row of $\mathbf{N}''$ — a contradiction.

Step 5. Induct on n . We prove (a) by induction on n . For $n = 1$, scale the single entry to be monic (or leave it 0). For $n \geq 2$, Steps 1–4 give a sequence of operations taking $\mathbf{M}$ to $f \oplus \mathbf{M}'$ with f monic and f dividing every entry of $\mathbf{M}'$. By the induction hypothesis there are operations taking $\mathbf{M}'$ to $\mathbf{D}' = \text{diag}(d_2, \dots, d_n)$ with every d_i monic or zero and $d_2 \mid \dots \mid d_n$; performing the corresponding operations on rows and columns 2, $\dots$, n of $f \oplus \mathbf{M}'$ takes it to $f \oplus \mathbf{D}' = \text{diag}(f, d_2, \dots, d_n)$. It remains to see that $f \mid d_2$. By Proposition 9.7.2, $\mathbf{D}' = \mathbf{U}'\mathbf{M}'\mathbf{V}'$ for some $\mathbf{U}', \mathbf{V}' \in M_{n-1}(F[x])$, so every entry of $\mathbf{D}'$ is a sum of products $u'_{ik}(\mathbf{M}')_{kl}v'_{lj}$ (Definition 0.9.9) and is therefore divisible by f , since every $(\mathbf{M}')_{kl}$ is. In particular $f \mid d_2$. Setting $d_1 = f$ finishes the induction.

- (b) Call a set $I \subseteq [n]$ with $|I| = k$ a **k -set**. For $\mathbf{M} \in M_n(F[x])$ and $1 \leq k \leq n$ define the **k -th determinantal divisor**

$$\Delta_k(\mathbf{M}) := \text{the monic generator of the ideal } \langle \det \mathbf{M}_{I,J} : I, J \text{ } k\text{-sets} \rangle \subseteq F[x],$$

and $\Delta_k(\mathbf{M}) := 0$ if all the $k \times k$ minors of $\mathbf{M}$ vanish. This is well defined by Proposition 5.2.1 and Theorem 5.2.2: the ideal generated by the finitely many $k \times k$ minors is $\langle \Delta_k(\mathbf{M}) \rangle$ for exactly one monic $\Delta_k(\mathbf{M})$ unless the ideal is $\{0\}$. Equivalently, $\Delta_k(\mathbf{M})$ is the monic greatest common divisor of the $k \times k$ minors: it divides each of them, and every common divisor of them divides it (Theorem 5.2.3, extended from two polynomials to finitely many by the same ideal argument).

Claim 1. $\mathbf{M} \approx \mathbf{M}'$ implies $\Delta_k(\mathbf{M}) = \Delta_k(\mathbf{M}')$ for every k .

Proof. Let $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$ with $\mathbf{U}, \mathbf{V}$ invertible over $F[x]$, and let I, J be k -sets. The submatrix of a product on rows I and columns J is the product of the submatrix of the left factor on rows I and all columns with the submatrix of the right factor on all rows and columns J (Definition 0.9.9, read off entry by entry). Applying this twice and then Theorem 6.7.2 over the commutative ring $F[x]$ — legitimate by the remark after that theorem — to the $k \times n$ times $n \times k$ products involved,

$$\det \mathbf{M}'_{I,J} = \sum_S \det \mathbf{U}_{I,S} \det (\mathbf{M}\mathbf{V})_{S,J} = \sum_S \sum_R \det \mathbf{U}_{I,S} \det \mathbf{M}_{S,R} \det \mathbf{V}_{R,J},$$

with S and R running over the k -sets. So every $k \times k$ minor of $\mathbf{M}'$ is an $F[x]$ -combination of $k \times k$ minors of $\mathbf{M}$, hence lies in $\langle \det \mathbf{M}_{S,R} \rangle$. Therefore $\Delta_k(\mathbf{M})$ divides every $k \times k$ minor of $\mathbf{M}'$, and so $\Delta_k(\mathbf{M}) \mid \Delta_k(\mathbf{M}')$. Exchanging the roles of $\mathbf{M}$ and $\mathbf{M}'$ using $\mathbf{M} = \mathbf{U}^{-1}\mathbf{M}'\mathbf{V}^{-1}$ gives $\Delta_k(\mathbf{M}') \mid \Delta_k(\mathbf{M})$. Two monic polynomials dividing each other are equal (Proposition 5.1.2), and if one is 0 so is the other. ■

Claim 2. If $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ with $d_1 \mid d_2 \mid \dots \mid d_n$, then $\Delta_k(\mathbf{D}) = d_1 d_2 \cdots d_k$.

Proof. Let I, J be k -sets. If $I \neq J$, pick $i \in I \setminus J$; row i of $\mathbf{D}$ has its only possibly non-zero entry in column $i \notin J$, so the submatrix $\mathbf{D}_{I,J}$ has a zero row and $\det \mathbf{D}_{I,J} = 0$. If $I = J = \{i_1 < \dots < i_k\}$, then $\mathbf{D}_{I,I} = \text{diag}(d_{i_1}, \dots, d_{i_k})$ and $\det \mathbf{D}_{I,I} = d_{i_1} \cdots d_{i_k}$ (Theorem 6.4.6, valid over the commutative ring $F[x]$ by the remark after it). Since $i_j \geq j$, the chain gives $d_j \mid d_{i_j}$ for each j , hence

$$d_1 d_2 \cdots d_k \mid d_{i_1} d_{i_2} \cdots d_{i_k}.$$

So $d_1 \cdots d_k$ divides every $k \times k$ minor of $\mathbf{D}$, and it is itself the minor for $I = J = \{1, \dots, k\}$. Hence the ideal generated by the $k \times k$ minors is $\langle d_1 \cdots d_k \rangle$. If $d_1 \cdots d_k \neq 0$ it is monic, being a product of monic polynomials, and equals $\Delta_k(\mathbf{D})$. If $d_1 \cdots d_k = 0$, say $d_m = 0$ with $m \leq k$, then for every k -set I we have $i_k \geq k \geq m$, so $d_m \mid d_{i_k}$ forces $d_{i_k} = 0$ and the minor vanishes; thus $\Delta_k(\mathbf{D}) = 0 = d_1 \cdots d_k$. ■

Now let $\mathbf{M} \approx \text{diag}(d_1, \dots, d_n)$ with the d_i as in (a). By Claims 1 and 2,

$$\Delta_k(\mathbf{M}) = d_1 d_2 \cdots d_k \quad (k = 1, \dots, n),$$

a quantity depending only on $\mathbf{M}$. Reading this backwards recovers the d_i : first $d_1 = \Delta_1(\mathbf{M})$; and for $k \geq 2$, if $\Delta_{k-1}(\mathbf{M}) \neq 0$ then $d_k = \Delta_k(\mathbf{M})/\Delta_{k-1}(\mathbf{M})$, while if $\Delta_{k-1}(\mathbf{M}) = 0$ then $d_1 \cdots d_{k-1} = 0$, so some $d_m = 0$ with $m \leq k-1$, and the chain forces $d_k = 0$. Hence any second list $(e_1, \dots, e_n)$ as in (a) satisfies the same recipe and equals $(d_1, \dots, d_n)$. This proves the theorem. ■

The proof is an algorithm, and it is the one to use by hand: hunt for the entry of least degree,

divide, and repeat.

Smith normal form of $\mathbf{M} \in M_n(F[x])$.

1. If $\mathbf{M} = 0$, stop. Otherwise find a non-zero entry of least degree and move it to position $(1, 1)$ by a row and a column swap; scale row 1 so that the entry is monic. Call it f .
2. For each other entry g in row 1 or column 1 with $f \nmid g$, replace g by its remainder on division by f (subtract a multiple of column 1, resp. row 1), move the remainder to the corner, and go back to step 1.
3. When f divides all of row 1 and column 1, clear them. If some entry g of the remaining block is not divisible by f , add its row to row 1 and go back to step 2.
4. Repeat on the remaining $(n - 1) \times (n - 1)$ block.
5. Check the answer: Δ_k must equal $d_1 \cdots d_k$. In particular $d_1 \cdots d_n$ is $\det \mathbf{M}$ rescaled by a non-zero constant to be monic, and d_1 is the monic gcd of the entries.

Quick check. Two polynomial matrices in $M_2(F[x])$ have determinants x^2 and x^2 , and in both the greatest common divisor of the four entries is 1. Are they equivalent over $F[x]$?

Solution. Yes. Here $\Delta_1 = 1$ (the gcd of the 1×1 minors, that is, of the entries) and Δ_2 is the determinant rescaled to be monic, so $\Delta_2 = x^2$ for both. By [Theorem 9.7.3](#) both have Smith normal form $\text{diag}(1, x^2)$, and since $\approx$ is an equivalence relation ([Proposition 9.7.2 \(a\)](#)) they are equivalent to each other. For 2×2 matrices the pair (Δ_1, Δ_2) is therefore a complete invariant.

Example. (A Smith Normal Form Computed by Hand)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Find the Smith normal form of $x\mathbf{I} - \mathbf{A}$ over $\mathbb{Q}[x]$, naming every operation, and check the answer with determinantal divisors.

Solution. We have

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-2 & -1 & 2 \\ -1 & x-2 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

Find the corner. The entries of least degree are the three entries equal to -1 . Swap columns 1 and 2, then multiply the first row by -1 :

$$\begin{pmatrix} -1 & x-2 & 2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -(x-2) & -2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

The corner is now 1, which divides everything, so no division step is needed.

Clear the first column. Add $-(x-2)$ times row 1 to row 2, and add row 1 to row 3. In row 2 this gives entries $-1 + (x-2)^2 = (x-1)(x-3)$ and $2 + 2(x-2) = 2(x-1)$; in

row 3 it gives $-1 - (x - 2) = -(x - 1)$ and $x + 1 - 2 = x - 1$:

$$\begin{pmatrix} 1 & -(x-2) & -2 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix}.$$

Clear the first row. Add $(x - 2)$ times column 1 to column 2, and 2 times column 1 to column 3. Since the first column below the corner is zero, nothing else changes:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix} = \text{diag}(1) \oplus \begin{pmatrix} (x-1)(x-3) & 2(x-1) \\ -(x-1) & x-1 \end{pmatrix}.$$

Every entry of the 2×2 block is divisible by 1, so we move on to the block.

The block. Its entries of least degree are the three of degree 1. Swap rows 2 and 3 and scale the new row 2 by -1 , putting the monic $x - 1$ in the corner of the block:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & -(x-1) \\ 0 & (x-1)(x-3) & 2(x-1) \end{pmatrix}.$$

Now $x - 1$ divides every entry of its row and column. Add column 2 to column 3, which clears the block's first row and turns the $(3, 3)$ -entry into $2(x - 1) + (x - 1)(x - 3) = (x - 1)(x - 1)$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & 0 \\ 0 & (x-1)(x-3) & (x-1)^2 \end{pmatrix}.$$

Add $-(x - 3)$ times row 2 to row 3 to clear the block's first column:

$$\mathbf{D} = \text{diag}(1, x-1, (x-1)^2),$$

and $1 \mid x - 1 \mid (x - 1)^2$, so this is the Smith normal form.

The check. The matrix $x\mathbf{I} - \mathbf{A}$ has an entry -1 , so $\Delta_1 = 1$. Its determinant is $p_{\mathbf{A}} = (x - 1)^3$ — expanding along the first row,

$$(x-2)[(x-2)(x+1)+2] + [-(x+1)+2] + 2[1+(x-2)] = (x-2)(x^2-x) + (1-x) + 2(x-1),$$

which is $(x-2)x(x-1) + (x-1) = (x-1)(x^2-2x+1) = (x-1)^3$ — so $\Delta_3 = (x-1)^3$. For Δ_2 , the minor on rows $\{2, 3\}$ and columns $\{1, 2\}$ is $\det \begin{pmatrix} -1 & x-2 \\ -1 & -1 \end{pmatrix} = 1 + (x-2) = x-1$, so $\Delta_2 \mid x - 1$; and every 2×2 minor is divisible by $x - 1$, as Claim 2 predicts from $\mathbf{D}$. Hence $\Delta_2 = x - 1$, and

$$(d_1, d_2, d_3) = \left(\Delta_1, \frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_2} \right) = (1, x-1, (x-1)^2),$$

in agreement. As a sanity check on $\mathbf{A}$ itself: $\mathbf{A} - \mathbf{I}$ has rank 1, so $g_{\mathbf{A}}(1) = 2$ and the Jordan form of $\mathbf{A}$ is $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1)$ (Corollary 9.3.2).

9.7.4 Similarity read off from a polynomial matrix

We can now state the theorem that makes all of this useful. It converts a question about $\mathbf{A}$ and $\mathbf{B}$ — is there an invertible $\mathbf{P} \in M_n(F)$ with $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$? — into a question about the two polynomial matrices $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{B}$, which the Smith algorithm answers by computation.

One direction is immediate. The other needs a tool: polynomial matrices can be divided by $x\mathbf{I} - \mathbf{A}$, with a **constant** remainder.

Lemma 9.7.4 — Division of a Polynomial Matrix by $x\mathbf{I} - \mathbf{A}$

Let $\mathbf{A} \in M_n(F)$ and $\mathbf{M} \in M_n(F[x])$. Then there are $\mathbf{Q} \in M_n(F[x])$ and $\mathbf{R} \in M_n(F)$ with

$$\mathbf{M} = (x\mathbf{I} - \mathbf{A})\mathbf{Q} + \mathbf{R}.$$

Idea. Write $\mathbf{M}$ as a polynomial $\sum_k \mathbf{M}_k x^k$ with **constant matrix** coefficients. The scalar identity $x^k - a^k = (x - a)(x^{k-1} + \cdots + a^{k-1})$ has a matrix version $x^k \mathbf{I} - \mathbf{A}^k = (x\mathbf{I} - \mathbf{A})\mathbf{S}_k$, in which the second factor is a polynomial in x and $\mathbf{A}$ and so commutes with $x\mathbf{I} - \mathbf{A}$. Replacing each $x^k \mathbf{I}$ by $(x^k \mathbf{I} - \mathbf{A}^k) + \mathbf{A}^k$ puts a factor $x\mathbf{I} - \mathbf{A}$ in front of everything except a constant matrix.

Proof. Collecting the coefficients of each power of x in the entries of $\mathbf{M}$, write $\mathbf{M} = \sum_{k=0}^m x^k \mathbf{M}_k$ with $\mathbf{M}_0, \dots, \mathbf{M}_m \in M_n(F)$; note that x^k is a scalar polynomial, so $x^k \mathbf{M}_k = \mathbf{M}_k x^k$. For $k \geq 1$ put

$$\mathbf{S}_k := \sum_{j=0}^{k-1} x^{k-1-j} \mathbf{A}^j \in M_n(F[x]), \quad \mathbf{S}_0 := 0.$$

Expanding the product and telescoping,

$$(x\mathbf{I} - \mathbf{A})\mathbf{S}_k = \sum_{j=0}^{k-1} x^{k-j} \mathbf{A}^j - \sum_{j=0}^{k-1} x^{k-1-j} \mathbf{A}^{j+1} = x^k \mathbf{I} - \mathbf{A}^k$$

for every $k \geq 0$. Therefore

$$\mathbf{M} = \sum_{k=0}^m (x^k \mathbf{I}) \mathbf{M}_k = \sum_{k=0}^m (x^k \mathbf{I} - \mathbf{A}^k) \mathbf{M}_k + \sum_{k=0}^m \mathbf{A}^k \mathbf{M}_k = (x\mathbf{I} - \mathbf{A}) \underbrace{\sum_{k=0}^m \mathbf{S}_k \mathbf{M}_k}_{=\mathbf{Q}} + \underbrace{\sum_{k=0}^m \mathbf{A}^k \mathbf{M}_k}_{=\mathbf{R}},$$

with $\mathbf{Q} \in M_n(F[x])$ and $\mathbf{R} \in M_n(F)$, as claimed. ■

Theorem 9.7.5 — Similarity as Equivalence of $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{B}$

Let F be a field, $n \geq 1$ and $\mathbf{A}, \mathbf{B} \in M_n(F)$. Then

$$\mathbf{A} \sim \mathbf{B} \text{ over } F \iff x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B} \text{ over } F[x].$$

Idea. $(\Rightarrow)$ is a one-line computation: a constant invertible $\mathbf{P}$ is also an invertible polynomial matrix.

$(\Leftarrow)$ is the substance, and the whole difficulty is that the matrices $\mathbf{U}, \mathbf{V}$ witnessing the equivalence have polynomial entries, while we must produce a **constant** $\mathbf{P}$. The trick is to divide $\mathbf{U}$ by $x\mathbf{I} - \mathbf{B}$ and keep only the constant remainder $\mathbf{P}$. Comparing degrees in the resulting identity shows that $\mathbf{P}$ alone already conjugates: $\mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{P}$, that is, $\mathbf{PA} = \mathbf{BP}$. Running the same argument on the inverted relation produces a second constant $\mathbf{P}'$ with $\mathbf{P}'\mathbf{B} = \mathbf{AP}'$, and multiplying the two expressions for $\mathbf{V}^{-1}$ and $\mathbf{V}$ shows $\mathbf{PP}' = \mathbf{I}$, so $\mathbf{P}$ is invertible.

Proof. $(\Rightarrow)$ Suppose $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$ with $\mathbf{P} \in M_n(F)$ invertible. Then, inside $M_n(F[x])$,

$$x\mathbf{I} - \mathbf{B} = \mathbf{P}^{-1}(x\mathbf{I})\mathbf{P} - \mathbf{P}^{-1}\mathbf{AP} = \mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P},$$

and $\mathbf{P}^{-1}, \mathbf{P} \in M_n(F) \subseteq M_n(F[x])$ have non-zero constant determinants, hence are invertible over $F[x]$ (Proposition 9.7.1). So $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$.

$(\Leftarrow)$ Suppose

$$x\mathbf{I} - \mathbf{B} = \mathbf{U}(x\mathbf{I} - \mathbf{A})\mathbf{V} \quad (*)$$

with $\mathbf{U}, \mathbf{V} \in M_n(F[x])$ invertible over $F[x]$.

Step 1. Divide $\mathbf{U}$ by $x\mathbf{I} - \mathbf{B}$. By Lemma 9.7.4 there are $\mathbf{Q} \in M_n(F[x])$ and $\mathbf{P} \in M_n(F)$ with $\mathbf{U} = (x\mathbf{I} - \mathbf{B})\mathbf{Q} + \mathbf{P}$. Substituting into $(*)$,

$$x\mathbf{I} - \mathbf{B} = (x\mathbf{I} - \mathbf{B})\mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{V} + \mathbf{P}(x\mathbf{I} - \mathbf{A})\mathbf{V},$$

so $\mathbf{P}(x\mathbf{I} - \mathbf{A})\mathbf{V} = (x\mathbf{I} - \mathbf{B})[\mathbf{I} - \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{V}]$. Multiplying on the right by $\mathbf{V}^{-1}$ and putting

$$\mathbf{S} := \mathbf{V}^{-1} - \mathbf{Q}(x\mathbf{I} - \mathbf{A}) \in M_n(F[x]),$$

we get

$$\mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{S}. \quad (1)$$

Step 2. $\mathbf{S}$ is a constant matrix, and $\mathbf{PA} = \mathbf{BP}$. Suppose $\mathbf{S} \neq 0$ and write $\mathbf{S} = \sum_{k=0}^d x^k \mathbf{S}_k$ with $\mathbf{S}_k \in M_n(F)$ and $\mathbf{S}_d \neq 0$. The coefficient of x^{d+1} in $(x\mathbf{I} - \mathbf{B})\mathbf{S}$ is $\mathbf{S}_d \neq 0$, while the left-hand side of (1) is $x\mathbf{P} - \mathbf{PA}$, whose coefficients vanish beyond x^1 . Hence $d+1 \leq 1$, that is $d = 0$. So in every case $\mathbf{S} \in M_n(F)$. Comparing the coefficients of x^1 in (1) gives $\mathbf{P} = \mathbf{S}$, and comparing the constant terms gives $-\mathbf{PA} = -\mathbf{BP}$. Therefore

$$\mathbf{PA} = \mathbf{BP}, \quad \text{equivalently} \quad \mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{P}, \quad (2)$$

the second form because $\mathbf{P}(x\mathbf{I} - \mathbf{A}) = x\mathbf{P} - \mathbf{PA} = x\mathbf{P} - \mathbf{BP} = (x\mathbf{I} - \mathbf{B})\mathbf{P}$. Moreover $\mathbf{S} = \mathbf{P}$ turns the definition of $\mathbf{S}$ into

$$\mathbf{V}^{-1} = \mathbf{P} + \mathbf{Q}(x\mathbf{I} - \mathbf{A}). \quad (3)$$

Step 3. The same argument, inverted. From $(*)$, $x\mathbf{I} - \mathbf{A} = \mathbf{U}^{-1}(x\mathbf{I} - \mathbf{B})\mathbf{V}^{-1}$, which is $(*)$ with $\mathbf{A}, \mathbf{B}$ interchanged and $\mathbf{U}, \mathbf{V}$ replaced by the invertible polynomial matrices $\mathbf{U}^{-1}, \mathbf{V}^{-1}$. So Steps 1 and 2 apply and give $\mathbf{Q}' \in M_n(F[x])$ and $\mathbf{P}' \in M_n(F)$ with

$$\mathbf{P}'(x\mathbf{I} - \mathbf{B}) = (x\mathbf{I} - \mathbf{A})\mathbf{P}' \quad \text{and} \quad \mathbf{V} = \mathbf{P}' + \mathbf{Q}'(x\mathbf{I} - \mathbf{B}). \quad (4)$$

Step 4. $\mathbf{P}$ is invertible. Multiplying (3) by (4),

$$\mathbf{I} = \mathbf{V}^{-1}\mathbf{V} = [\mathbf{P} + \mathbf{Q}(x\mathbf{I} - \mathbf{A})][\mathbf{P}' + \mathbf{Q}'(x\mathbf{I} - \mathbf{B})] = \mathbf{P}\mathbf{P}' + \mathbf{P}\mathbf{Q}'(x\mathbf{I} - \mathbf{B}) + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{P}' + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{Q}'(x\mathbf{I} - \mathbf{B}).$$

By the first identity in (4), $(x\mathbf{I} - \mathbf{A})\mathbf{P}' = \mathbf{P}'(x\mathbf{I} - \mathbf{B})$, so the third summand is $\mathbf{Q}\mathbf{P}'(x\mathbf{I} - \mathbf{B})$ and

$$\mathbf{I} - \mathbf{P}\mathbf{P}' = \mathbf{Y}(x\mathbf{I} - \mathbf{B}), \quad \mathbf{Y} := \mathbf{P}\mathbf{Q}' + \mathbf{Q}\mathbf{P}' + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{Q}' \in M_n(F[x]).$$

If $\mathbf{Y} \neq 0$, write $\mathbf{Y} = \sum_{k=0}^e x^k \mathbf{Y}_k$ with $\mathbf{Y}_e \neq 0$; the coefficient of x^{e+1} on the right is $\mathbf{Y}_e \neq 0$, while the left-hand side is constant. Hence $\mathbf{Y} = 0$ and $\mathbf{P}\mathbf{P}' = \mathbf{I}$. By Theorem 2.4.5, $\mathbf{P}$ is invertible with $\mathbf{P}^{-1} = \mathbf{P}'$.

Step 5. Conclude. By (2), $\mathbf{P}\mathbf{A} = \mathbf{B}\mathbf{P}$, so $\mathbf{B} = \mathbf{P}\mathbf{A}\mathbf{P}^{-1}$ and $\mathbf{A} \sim \mathbf{B}$. This proves the theorem. ■

Two by-products are worth recording. Whenever $x\mathbf{I} - \mathbf{B} = \mathbf{U}(x\mathbf{I} - \mathbf{A})\mathbf{V}$ with $\mathbf{U}, \mathbf{V}$ invertible over $F[x]$, taking determinants gives $p_{\mathbf{B}} = (\det \mathbf{U})(\det \mathbf{V})p_{\mathbf{A}}$ with $\det \mathbf{U}, \det \mathbf{V}$ non-zero constants (Proposition 9.7.1); both $p_{\mathbf{A}}$ and $p_{\mathbf{B}}$ are monic of degree n , so

$$p_{\mathbf{A}} = p_{\mathbf{B}} \quad \text{and} \quad (\det \mathbf{U})(\det \mathbf{V}) = 1,$$

which recovers the invariance of the characteristic polynomial. And, by Theorem 9.7.3, the test is now decidable by a finite computation in $F[x]$: reduce $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{B}$ to Smith normal form and compare.

Corollary 9.7.6 — The Smith Form of $x\mathbf{I} - \mathbf{A}$ Carries the Invariant Factors

Let $\mathbf{A} \in M_n(F)$ with invariant factors $d_1 \mid d_2 \mid \cdots \mid d_r$ as in Theorem 9.6.5. Then the Smith normal form of $x\mathbf{I} - \mathbf{A}$ is

$$\text{diag} \left(\underbrace{1, \dots, 1}_{n-r}, d_1, d_2, \dots, d_r \right).$$

In particular the invariant factors of $\mathbf{A}$ are exactly the non-constant diagonal entries of the Smith normal form of $x\mathbf{I} - \mathbf{A}$, read from the top down, and two matrices in $M_n(F)$ are similar if and only if $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{B}$ have the same Smith normal form.

Proof. By Theorem 9.6.5, $\mathbf{A} \sim \mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$ with d_i monic, $\deg d_i = k_i \geq 1$ and $\sum k_i = n$. By the ($\Rightarrow$) direction of Theorem 9.7.5 it is enough to find the Smith normal form of

$$x\mathbf{I}_n - (\mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)) = (x\mathbf{I}_{k_1} - \mathbf{C}(d_1)) \oplus \cdots \oplus (x\mathbf{I}_{k_r} - \mathbf{C}(d_r)),$$

the equality by Theorem 7.1.4 (a).

One block. Let d be monic of degree $k \geq 1$. By Definition 9.5.3, the only non-zero entries of $x\mathbf{I}_k - \mathbf{C}(d)$ are x on the diagonal (with $x + a_{k-1}$ in the corner), -1 in each position $(i+1, i)$, and the entries of the last column. Delete row 1 and column k . In the remaining $(k-1) \times (k-1)$ matrix, the entry in position (p, q) is the $(p+1, q)$ -entry of $x\mathbf{I}_k - \mathbf{C}(d)$: it is -1 when $q = p$, it is x when $q = p+1$, and it is 0 when $q < p$.

So that matrix is upper triangular with every diagonal entry -1 , and its determinant is $(-1)^{k-1} \neq 0$ by Theorem 6.4.6, again over $F[x]$. Hence $\Delta_{k-1}(x\mathbf{I}_k - \mathbf{C}(d)) = 1$. By Claim 2 in the proof of Theorem 9.7.3, the determinantal divisors of any matrix are the partial products $d_1 \cdots d_j$ of its Smith diagonal, so $\Delta_j \mid \Delta_{k-1} = 1$ and therefore $\Delta_j = 1$ for every $j \leq k-1$. Also $\Delta_k = \det(x\mathbf{I}_k - \mathbf{C}(d)) = p_{\mathbf{C}(d)} = d$ by Theorem 9.5.3. By the uniqueness part of Theorem 9.7.3, the Smith normal form of $x\mathbf{I}_k - \mathbf{C}(d)$ is $\text{diag}(1, \dots, 1, d)$. (For $k = 1$ read this as $\text{diag}(d)$.)

Assembling. If $\mathbf{M}_i \approx \mathbf{M}'_i$ with witnesses $\mathbf{U}_i, \mathbf{V}_i$, then $\bigoplus_i \mathbf{M}_i \approx \bigoplus_i \mathbf{M}'_i$ with witnesses $\bigoplus_i \mathbf{U}_i$ and $\bigoplus_i \mathbf{V}_i$, which are invertible with inverses $\bigoplus_i \mathbf{U}_i^{-1}, \bigoplus_i \mathbf{V}_i^{-1}$ (Theorem 7.1.4 (a), (c)). So $x\mathbf{I}_n - \mathbf{A}$ is equivalent to the direct sum of the matrices $\text{diag}(1, \dots, 1, d_i)$, a diagonal matrix whose diagonal entries are $n - r$ ones and the polynomials $d_1, \dots, d_r$. Simultaneous row and column swaps — each a pair of elementary operations — reorder these entries at will, so $x\mathbf{I}_n - \mathbf{A}$ is equivalent to

$$\text{diag}(1, \dots, 1, d_1, d_2, \dots, d_r),$$

and this is already in the shape required by Theorem 9.7.3 (a), because $1 \mid d_1 \mid d_2 \mid \cdots \mid d_r$ and all the entries are monic. By uniqueness it *is* the Smith normal form of $x\mathbf{I} - \mathbf{A}$. The final claim combines this with Theorem 9.7.5 and the uniqueness in Theorem 9.7.3. ■

So the determinantal divisors of $x\mathbf{I} - \mathbf{A}$ compute the invariant factors of $\mathbf{A}$ without any factorization of polynomials: Δ_k is a gcd, obtained from the Euclidean algorithm alone. Moreover $\Delta_n(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}$ and $\Delta_n/\Delta_{n-1} = d_r = m_{\mathbf{A}}$.

9.7.5 Every matrix is similar to its transpose

Chapter 3 asked whether $\mathbf{A}$ and $\mathbf{A}^\top$ are always similar and postponed the answer; Exercise (C1: A matrix is similar to its transpose) settled it whenever $p_{\mathbf{A}}$ splits. The Smith form removes that hypothesis, because transposing a matrix does not change its minors.

Corollary 9.7.7 — A Matrix Is Similar to Its Transpose

Let F be any field and $\mathbf{A} \in M_n(F)$. Then $\mathbf{A} \sim \mathbf{A}^\top$.

Proof. Let $\mathbf{M} = x\mathbf{I} - \mathbf{A}$, so that $\mathbf{M}^\top = x\mathbf{I} - \mathbf{A}^\top$. For k -sets I, J we have $(\mathbf{M}^\top)_{I,J} = (\mathbf{M}_{J,I})^\top$ by Definition 0.9.10, so $\det(\mathbf{M}^\top)_{I,J} = \det \mathbf{M}_{J,I}$ by Theorem 6.4.5, valid over the commutative ring $F[x]$. Hence the $k \times k$ minors of $\mathbf{M}^\top$ are exactly the $k \times k$ minors of $\mathbf{M}$, listed in a different order, and

$$\Delta_k(x\mathbf{I} - \mathbf{A}^\top) = \Delta_k(x\mathbf{I} - \mathbf{A}) \quad (k = 1, \dots, n).$$

By Theorem 9.7.3, each of $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{A}^\top$ is equivalent over $F[x]$ to the diagonal matrix built from its own determinantal divisors, and these two diagonal matrices coincide. Since $\approx$ is an equivalence relation (Proposition 9.7.2 (a)), $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{A}^\top$. By Theorem 9.7.5, $\mathbf{A} \sim \mathbf{A}^\top$. ■

Two sentences of consequence. An operator and its dual have the same invariants: by Theorem 4.4.2 the matrix of T' in the dual basis is $[T]_{\mathcal{B}}^{\top}$, so T and T' share their characteristic polynomial, their minimal polynomial and their invariant factors. And a matrix and its transpose always have the same Jordan form when one exists, which says that the *rows* of $\mathbf{A} - \lambda\mathbf{I}$ are exactly as informative as its columns.

Warning. **Similar is not the same as “equivalent over $F[x]$ ” for the matrices themselves.** The theorem compares $x\mathbf{I} - \mathbf{A}$ with $x\mathbf{I} - \mathbf{B}$, not $\mathbf{A}$ with $\mathbf{B}$. As polynomial matrices, any two invertible $\mathbf{A}, \mathbf{B} \in M_n(F)$ are equivalent over $F[x]$ — take $\mathbf{U} = \mathbf{A}^{-1}$, $\mathbf{V} = \mathbf{B}$ — whether or not they are similar. The whole content of Theorem 9.7.5 is that inserting the variable x on the diagonal remembers the conjugation.

9.7.6 Exercises

9.7.6.1 A. Check your understanding

Exercise A1.

- List the three elementary operations over $F[x]$, and say which one differs from its counterpart over a field and why.
- State a criterion for a matrix $\mathbf{U} \in M_n(F[x])$ to be invertible in $M_n(F[x])$.
- State the Smith Normal Form theorem, including the uniqueness clause.
- Define the k -th determinantal divisor of a polynomial matrix, and say why it is unchanged by equivalence.
- True or false: if $\det(x\mathbf{I} - \mathbf{A}) = \det(x\mathbf{I} - \mathbf{B})$, then $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$. Justify your answer.
- How do you read $m_{\mathbf{A}}$ off the Smith normal form of $x\mathbf{I} - \mathbf{A}$?

Solution.

- Swap two rows; multiply a row by a non-zero **scalar** of F ; add a **polynomial** multiple of one row to a different row (Definition 9.7.1). The scaling operation is restricted, because only the non-zero constants are invertible in $F[x]$; multiplying a row by x is not reversible.
- $\mathbf{U}$ is invertible over $F[x]$ if and only if $\det \mathbf{U}$ is a non-zero constant (Proposition 9.7.1).
- Every $\mathbf{M} \in M_n(F[x])$ is equivalent over $F[x]$, by elementary operations, to a diagonal $\text{diag}(d_1, \dots, d_n)$ with each d_i monic or zero and $d_i \mid d_{i+1}$; this diagonal is unique (Theorem 9.7.3).
- $\Delta_k(\mathbf{M})$ is the monic generator of the ideal generated by all $k \times k$ minors of $\mathbf{M}$, equivalently their monic gcd. It is unchanged by equivalence because every $k \times k$ minor of $\mathbf{UMV}$ is an $F[x]$ -combination of $k \times k$ minors of $\mathbf{M}$, by the Cauchy–Binet formula (Claim 1 in the proof of Theorem 9.7.3).
- False. Equality of determinants says only $\Delta_n(x\mathbf{I} - \mathbf{A}) = \Delta_n(x\mathbf{I} - \mathbf{B})$; the lower determinantal divisors can differ. For $\mathbf{A} = \mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$ and $\mathbf{B} = \mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$ in $M_4(F)$, both determinants are x^4 , but the Smith normal forms are $\text{diag}(1, 1, x^2, x^2)$ and $\text{diag}(1, 1, x, x^3)$. (Equivalently, the matrices are not similar, by Corollary 9.3.3.)

- (f) $m_{\mathbf{A}}$ is the **last** diagonal entry of the Smith normal form of $x\mathbf{I} - \mathbf{A}$, that is $\Delta_n/\Delta_{n-1} = p_{\mathbf{A}}/\Delta_{n-1}$ (Corollary 9.7.6, since the last diagonal entry is $d_r = m_{\mathbf{A}}$ in the notation there).

9.7.6.2 B. Practice

Exercise B1: Smith normal forms.

Find the Smith normal form over $\mathbb{Q}[x]$ of each of the following, naming the operations you use.

- (a) $\mathbf{M}_1 = \begin{pmatrix} x & x+1 \\ x-1 & x \end{pmatrix}$.
- (b) $\mathbf{M}_2 = \begin{pmatrix} x^2-1 & 0 \\ 0 & x^2-x \end{pmatrix}$.
- (c) $\mathbf{M}_3 = \begin{pmatrix} x & 0 & 1 \\ -1 & x & 0 \\ 0 & -1 & x \end{pmatrix}$.

Solution.

- (a) $\det \mathbf{M}_1 = x^2 - (x+1)(x-1) = 1$, so $\mathbf{M}_1$ is invertible over $\mathbb{Q}[x]$ (Proposition 9.7.1) and $\Delta_1 = \Delta_2 = 1$. The Smith normal form is $\text{diag}(1, 1) = \mathbf{I}_2$. (Concretely: subtract column 1 from column 2 to get $\begin{pmatrix} x & 1 \\ x-1 & 1 \end{pmatrix}$, then subtract row 1 from row 2, swap columns, and clear.)
- (b) Here $\Delta_1 = \gcd(x^2-1, x^2-x) = \gcd((x-1)(x+1), x(x-1)) = x-1$ and $\Delta_2 = \det \mathbf{M}_2 = x(x-1)^2(x+1)$. So the Smith normal form is

$$\text{diag}\left(x-1, \frac{x(x-1)^2(x+1)}{x-1}\right) = \text{diag}(x-1, x(x-1)(x+1)),$$

and indeed $x-1$ divides $x(x-1)(x+1)$. By operations: add column 2 to column 1, giving first column (x^2-1, x^2-x) ; subtract row 1 from row 2 to get $(x^2-1, 1-x)$ in column 1; the entry $1-x$ has least degree, so swap it into the corner, scale by -1 , and continue.

- (c) The 2×2 minor on rows $\{2, 3\}$ and columns $\{1, 2\}$ is $\det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} = 1$, so $\Delta_2 = 1$, and then $\Delta_1 = 1$ as well (it divides Δ_2). Expanding $\det \mathbf{M}_3$ along the first row, $x(x^2-0) + 1 \cdot (1-0) = x^3+1$, so $\Delta_3 = x^3+1$ and the Smith normal form is $\text{diag}(1, 1, x^3+1)$. Note $\mathbf{M}_3 = x\mathbf{I} - \mathbf{C}$ with $\mathbf{C} = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \mathbf{C}(x^3+1)$, and Corollary 9.7.6 predicts exactly this, since a companion matrix has the single invariant factor x^3+1 .

Exercise B2: Invariant factors from determinantal divisors.

For each matrix $\mathbf{A}$, compute $\Delta_1, \dots, \Delta_n$ of $x\mathbf{I} - \mathbf{A}$, write down the Smith normal form, and read off the invariant factors, $m_{\mathbf{A}}$ and the Jordan form.

- (a) $\mathbf{A} = \mathbf{J}_2(0) \oplus \mathbf{J}_2(0) \in M_4(\mathbb{Q})$.
 (b) $\mathbf{A} = \mathbf{J}_3(0) \oplus \mathbf{J}_1(0) \in M_4(\mathbb{Q})$.
 (c) $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{Q})$.

Solution. One remark makes the minors easy. If $\mathbf{M} = \mathbf{M}_1 \oplus \mathbf{M}_2$ with square blocks, and I, J are k -sets, split $I = I_1 \cup I_2$ and $J = J_1 \cup J_2$ according to the blocks. If $|I_1| \neq |J_1|$, then $\mathbf{M}_{I,J}$ has a zero row or a zero column and $\det \mathbf{M}_{I,J} = 0$; if $|I_1| = |J_1|$, then $\mathbf{M}_{I,J} = (\mathbf{M}_1)_{I_1,J_1} \oplus (\mathbf{M}_2)_{I_2,J_2}$ and the determinant is the product of the two, by Theorem 6.8.4, whose proof uses only sums and products of entries and so applies over $F[x]$ (the remark after that theorem). So **the minors of a block diagonal matrix are the products of matching-size minors of the blocks**. Here $x\mathbf{I} - \mathbf{A}$ is the direct sum of the blocks $x\mathbf{I}_k - \mathbf{J}_k(0)$, each upper triangular with x on the diagonal and -1 on the superdiagonal.

- (a) $x\mathbf{I} - \mathbf{A} = \mathbf{N} \oplus \mathbf{N}$ with $\mathbf{N} = \begin{pmatrix} x & -1 \\ 0 & x \end{pmatrix}$, whose minors are the entries $x, -1, 0$ for $k = 1$ and x^2 for $k = 2$. By the remark, the minors of $\mathbf{N} \oplus \mathbf{N}$ are $\{x, -1, 0\}$ for $k = 1$; the products of two 1×1 minors together with x^2 for $k = 2$; x^2 times a 1×1 minor for $k = 3$; and x^4 for $k = 4$. Hence

$$\Delta_1 = 1, \quad \Delta_2 = 1 \quad (\text{from } (-1)(-1) = 1), \quad \Delta_3 = x^2, \quad \Delta_4 = x^4,$$

so the Smith normal form is $\text{diag}(1, 1, x^2, x^2)$, the invariant factors are $x^2 \mid x^2$, $m_{\mathbf{A}} = x^2$ is the last of them, and the Jordan form is $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$.

- (b) Now $x\mathbf{I} - \mathbf{A} = \mathbf{N}_3 \oplus (x)$ with $\mathbf{N}_3 = \begin{pmatrix} x & -1 & 0 \\ 0 & x & -1 \\ 0 & 0 & x \end{pmatrix}$. The minors of $\mathbf{N}_3$ are $\{x, -1, 0\}$ for $k = 1$; $\{x^2, -x, 1, 0\}$ for $k = 2$, the value 1 coming from rows $\{1, 2\}$ and columns $\{2, 3\}$; and x^3 for $k = 3$. By the remark, the $k \times k$ minors of $x\mathbf{I} - \mathbf{A}$ are those of $\mathbf{N}_3$ together with x times its $(k-1) \times (k-1)$ minors, so

$$\Delta_1 = 1, \quad \Delta_2 = \gcd(x^2, -x, 1, x \cdot x, -x) = 1, \quad \Delta_3 = \gcd(x^3, x \cdot x^2, -x \cdot x, x \cdot 1) = x, \quad \Delta_4 = x^4.$$

The Smith normal form is $\text{diag}(1, 1, x, x^3)$: the invariant factors are $x \mid x^3$, $m_{\mathbf{A}} = x^3$ is the last of them, and the Jordan form is $\mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$. Compared with (a) the determinants agree but Δ_3 does not, which is exactly why the two matrices are not similar.

- (c) $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-3 & -1 \\ 0 & x-3 \end{pmatrix}$. Then $\Delta_1 = 1$ and $\Delta_2 = (x-3)^2$, so the Smith normal form is $\text{diag}(1, (x-3)^2)$, the only invariant factor is $(x-3)^2 = m_{\mathbf{A}} = p_{\mathbf{A}}$, and the Jordan form is $\mathbf{J}_2(3)$. Consistently, $\mathbf{A}$ is a companion matrix up to similarity, since $m_{\mathbf{A}} = p_{\mathbf{A}}$ (Theorem 9.5.6).

Exercise B3: Deciding similarity.

Determine whether each pair is similar over the stated field. Justify your answer using Theorem 9.7.5.

- (a) $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ over $\mathbb{Q}$.
- (b) $\mathbf{A} = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}$ and $\mathbf{B} = \mathbf{J}_2(1) \oplus \mathbf{J}_1(1)$ over $\mathbb{Q}$.
- (c) $\mathbf{A} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$ over $\mathbb{Q}$.

Solution.

- (a) Similar. Here $\mathbf{B} = \mathbf{A}^\top$, so [Corollary 9.7.7](#) applies. (Explicitly, $\Delta_1 = 1$ and $\Delta_2 = x^2 + 1$ for both, so both Smith normal forms are $\text{diag}(1, x^2 + 1)$.)
- (b) Similar. By [Example \(A Smith Normal Form Computed by Hand\)](#) the Smith normal form of $x\mathbf{I} - \mathbf{A}$ is $\text{diag}(1, x-1, (x-1)^2)$. For $\mathbf{B}$, $x\mathbf{I} - \mathbf{B} = \begin{pmatrix} x-1 & -1 \\ 0 & x-1 \end{pmatrix} \oplus (x-1)$. Its entry -1 gives $\Delta_1 = 1$. By the block-minor rule in the solution to [Exercise \(B2: Invariant factors from determinantal divisors\)](#), the 2×2 minors are $(x-1)^2$, from the 2×2 block, together with the products of a 1×1 minor of each block, that is $(x-1)^2$, $-(x-1)$ and 0 ; so $\Delta_2 = x-1$. Finally $\Delta_3 = \det(x\mathbf{I} - \mathbf{B}) = (x-1)^3$. The Smith normal forms agree, so $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$ and $\mathbf{A} \sim \mathbf{B}$.
- (c) Similar. Here

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x & 2 \\ -1 & x \end{pmatrix}, \quad x\mathbf{I} - \mathbf{B} = \begin{pmatrix} x & 1 \\ -2 & x \end{pmatrix},$$

and each has an entry that is a non-zero constant, so $\Delta_1 = 1$ in both cases; and $\Delta_2 = \det(x\mathbf{I} - \mathbf{A}) = x^2 + 2 = \det(x\mathbf{I} - \mathbf{B})$. So both Smith normal forms are $\text{diag}(1, x^2 + 2)$, hence $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$ and $\mathbf{A} \sim \mathbf{B}$ by [Theorem 9.7.5](#). Note that $x^2 + 2$ is irreducible over $\mathbb{Q}$, so no Jordan form exists over $\mathbb{Q}$ and Exercise (C1: A matrix is similar to its transpose) would not have decided this pair. (Here $\mathbf{B} \neq \mathbf{A}^\top$, so [Corollary 9.7.7](#) does not apply directly; the Smith forms settle it anyway.)

9.7.6.3 C. Going deeper

Exercise C1: The same algorithm over the integers.

The proof of [Theorem 9.7.3](#) used only one property of $F[x]$: division with remainder, where the “size” of a non-zero element is its degree. The integers $\mathbb{Z}$ have the same property, with $|a|$ in place of $\deg$.

- (a) State the version of [Theorem 9.7.3](#) for $\mathbf{M} \in M_n(\mathbb{Z})$ that the same proof gives, saying what replaces “monic”.
- (b) Compute the Smith normal form over $\mathbb{Z}$ of $\mathbf{M} = \begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix}$, naming the operations.
- (c) Every finite abelian group G with n generators is $\mathbb{Z}^n$ modulo the column span of some $\mathbf{M} \in M_n(\mathbb{Z})$, the matrix of relations, exactly as V was $F[x]^n$ modulo the columns of $x\mathbf{I} - \mathbf{A}$. Using (b), identify the group with relation matrix $\mathbf{M}$ as a product of cyclic groups.

Hint: for (c), a diagonal relation matrix $\text{diag}(d_1, \dots, d_n)$ presents the product of the

cyclic groups $\mathbb{Z}/d_i\mathbb{Z}$.

Solution.

- (a) Every $\mathbf{M} \in M_n(\mathbb{Z})$ can be brought by integer elementary operations — swap two rows or columns, multiply a row or column by ± 1 (the units of $\mathbb{Z}$), add an **integer** multiple of one row or column to another — to a diagonal matrix $\text{diag}(d_1, \dots, d_n)$ with every $d_i \geq 0$ and $d_i \mid d_{i+1}$, and this diagonal is unique. “Monic” is replaced by “non-negative”, which is the normalization available because the units of $\mathbb{Z}$ are ± 1 . Uniqueness again follows from the determinantal divisors Δ_k , now non-negative integer gcds of the $k \times k$ minors, using the Cauchy–Binet formula over the commutative ring $\mathbb{Z}$.
- (b) The least absolute value among the entries is 2, already in the corner. It divides 4 and 6. Subtract 2 times column 1 from column 2, and 3 times row 1 from row 2:

$$\begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 6 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix},$$

the last step multiplying row 2 by -1 . Since $2 \mid 8$, the Smith normal form is $\text{diag}(2, 8)$. Check: $\Delta_1 = \gcd(2, 4, 6, 4) = 2$ and $\Delta_2 = |\det \mathbf{M}| = |8 - 24| = 16 = 2 \cdot 8$.

- (c) By (b) and the hint, the group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, of order $16 = |\det \mathbf{M}|$. This is the structure theorem for finite abelian groups in miniature: the invariant divisors $d_1 \mid \dots \mid d_n$ of the relation matrix are the orders of the cyclic factors.

Exercise C2: Uniqueness by hand in the smallest case.

Let $\mathbf{M} \in M_2(F[x])$ be non-zero.

- (a) Prove directly, without using determinantal divisors, that $\Delta_1(\mathbf{M})$ — the monic gcd of the four entries of $\mathbf{M}$ — is unchanged by each of the six elementary operations of [Definition 9.7.1](#).
- (b) Prove that $\det(\mathbf{U}\mathbf{M}\mathbf{V}) = c \det \mathbf{M}$ for a non-zero constant c whenever $\mathbf{U}, \mathbf{V}$ are invertible over $F[x]$.
- (c) Deduce that if $\text{diag}(d_1, d_2) \approx \text{diag}(e_1, e_2)$ with all four entries monic or zero, $d_1 \mid d_2$ and $e_1 \mid e_2$, then $d_1 = e_1$ and $d_2 = e_2$.

Solution.

- (a) Write $d = \Delta_1(\mathbf{M})$, the monic gcd of the entries, that is, the monic generator of the ideal $\langle m_{11}, m_{12}, m_{21}, m_{22} \rangle$ (Theorem 5.2.2). A swap permutes the entries, so the ideal is unchanged. Scaling a row by $c \in F \setminus \{0\}$ replaces two generators g by cg , and $\langle cg \rangle = \langle g \rangle$ since c is a unit, so the ideal is unchanged. Adding h times row 1 to row 2 replaces m_{2j} by $m_{2j} + hm_{1j}$, which lies in the old ideal; so the new ideal is contained in the old one, and the operation is reversed by adding $-h$ times row 1 to row 2, which gives the reverse containment. The same three arguments apply to columns. Hence d is unchanged.
- (b) By multiplicativity of the determinant over the commutative ring $F[x]$ (Theorem 6.5.4 and the remark on commutative rings), $\det(\mathbf{U}\mathbf{M}\mathbf{V}) = \det \mathbf{U} \cdot \det \mathbf{M} \cdot \det \mathbf{V}$, and $\det \mathbf{U}, \det \mathbf{V}$ are non-zero constants by [Proposition 9.7.1](#). Take $c = \det \mathbf{U} \det \mathbf{V}$.

- (c) By [Proposition 9.7.2](#) and (a), Δ_1 is an invariant of equivalence as far as elementary operations go; and for a general equivalence $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$ the argument of (a) applies verbatim with the generators of the ideal replaced using $m'_{ij} = \sum_{k,l} u_{ik}m_{kl}v_{lj} \in \langle m_{kl} \rangle$, and symmetrically. For the two diagonal matrices, $\Delta_1(\text{diag}(d_1, d_2)) = \gcd(d_1, d_2) = d_1$ since $d_1 \mid d_2$ and d_1 is monic (or $d_1 = d_2 = 0$); likewise $\Delta_1(\text{diag}(e_1, e_2)) = e_1$. Hence $d_1 = e_1$. By (b), $d_1d_2 = ce_1e_2$ for some non-zero constant c ; if $d_1 = e_1 \neq 0$ then $d_2 = ce_2$, and two monic polynomials (or two zeros) differing by a constant factor are equal, so $d_2 = e_2$. If $d_1 = e_1 = 0$, then $d_2 = e_2 = 0$ by the divisibility chain.